Automotive Data Analysis: A SQL Study by Landon Mispagel

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, I am sent an array of questions that are being asked at the leadership level that need to be answered using SQL. The tables used for the project are not included in this google doc, only the outputs.

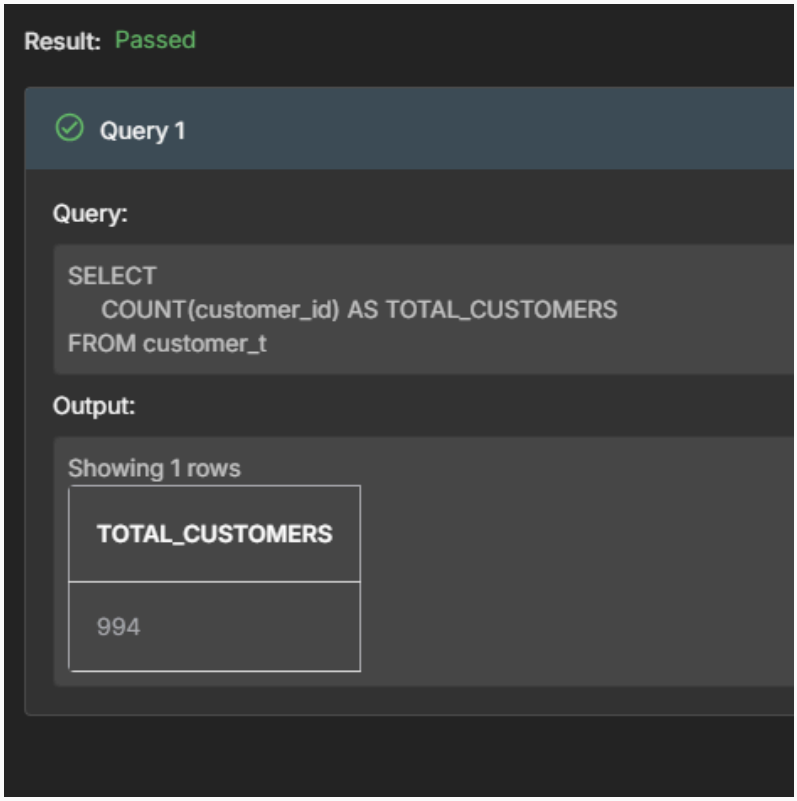
Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query: A

```
Total Customers:
   1. SELECT
   2.      COUNT(customer_id) AS TOTAL_CUSTOMERS
   3. FROM customer_t ;
```

Output:



Solution Query: B

```
Distribution by state:
SELECT
     state,
     COUNT(customer_id) AS TOTAL_CUSTOMERS
FROM customer_t
GROUP BY state ;
```

Result: Passed

Query 1

Query:

```
SELECT
    state,
    COUNT(customer_id) AS TOTAL_CUSTOMERS
FROM customer_t
GROUP BY state
ORDER BY TOTAL_CUSTOMERS DESC
```

Output:

Showing first 10 rows out of 49 rows

state	TOTAL_CUSTOMERS
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33

Observations and Insights:

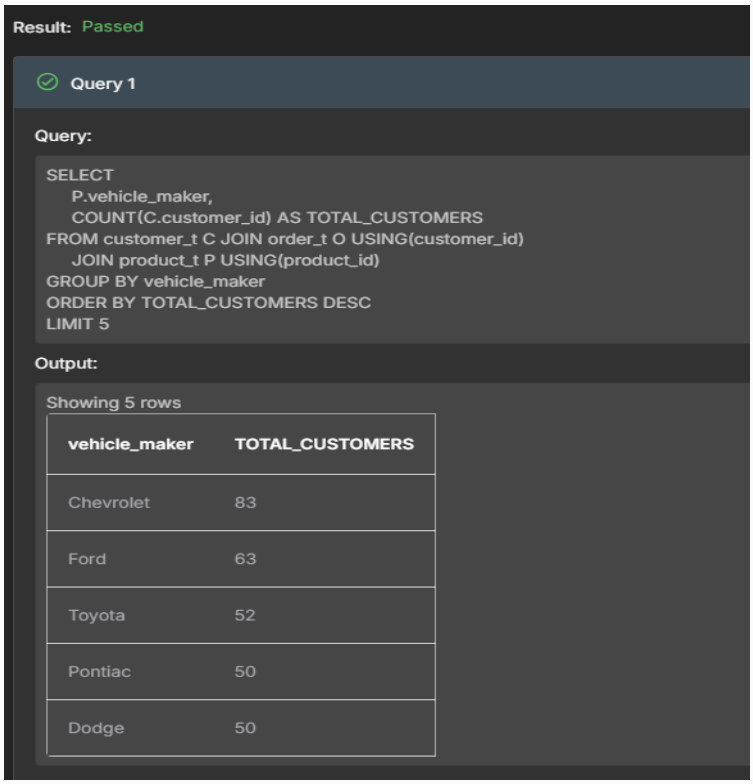
- The total customer count is 994.
- The most customers reside in Texas and California with 97 customers each.
- The least customers reside in Maine, Vermont, and Wyoming with 1 customer.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
      P.vehicle_maker,
      COUNT(C.customer_id) AS TOTAL_CUSTOMERS
FROM customer_t C JOIN order_t O USING(customer_id)
      JOIN product_t P USING(product_id)
GROUP BY vehicle_maker
ORDER BY TOTAL_CUSTOMERS DESC
LIMIT 5 ;
```

Output:



vehicle_maker	TOTAL_CUSTOMERS
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- The top five vehicle makers preferred are Chevrolet, Ford, Toyota, Pontiac, and Dodge.
- Chevrolet is the most popular with a total of 83 customers.
- Pontiac and Dodge have an equal customer population with 50 each.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
     C.state,
     P.vehicle_maker,
FROM customer_t C JOIN order_t O USING(customer_id)
     JOIN product_t P USING(product_id)
GROUP BY C.state
ORDER BY COUNT(C.customer_id) DESC;
```

Output:

Output:

Showing first 10 rows out of 49 rows

state	vehicle_maker
Texas	Chevrolet
California	Toyota
Florida	GMC
New York	Plymouth
District of Columbia	Dodge
Ohio	Mazda

Observations and Insights:

- The most popular vehicle maker in Alabama.
- Cadillac and Pontiac are tied for Rank 1 with 3 customers each.
- Chevrolet is the most popular vehicle maker in Texas with a total of 9 customers.

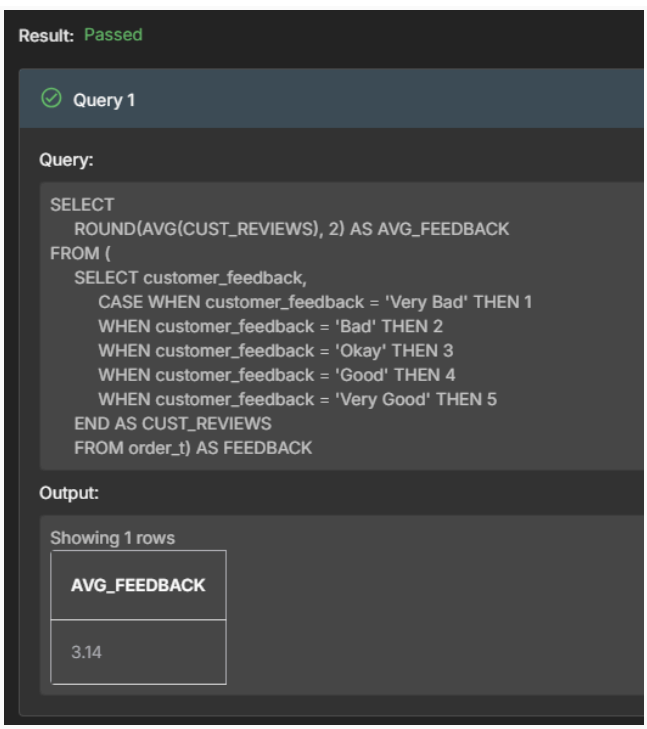
Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query: A

```
Average Feedback Total:
SELECT
     ROUND(AVG(CUST_REVIEWS), 2) AS AVG_FEEDBACK
FROM (
     SELECT customer_feedback,
          CASE WHEN customer_feedback = ‘Very Bad’ THEN 1
          WHEN customer_feedback = ‘Bad’ THEN 2
          WHEN customer_feedback = ‘Okay’ THEN 3
          WHEN customer_feedback = ‘Good’ THEN 4
          WHEN customer_feedback = ‘Very Good’ THEN 5
       END AS CUST_REVIEWS
       FROM order_t) AS FEEDBACK ;
```

Output:



Solution Query: B

```
Average Feedback Per Quarter:
SELECT
     ROUND(AVG(CUST_REVIEWS), 2) AS AVG_FEEDBACK,
     quarter_number
FROM (
    SELECT customer_feedback, quarter_number,
        CASE WHEN customer_feedback = 'Very Bad' THEN 1
        WHEN customer_feedback = 'Bad' THEN 2
        WHEN customer_feedback = 'Okay' THEN 3
        WHEN customer_feedback = 'Good' THEN 4
           WHEN customer_feedback = 'Very Good' THEN 5
       END AS CUST_REVIEWS
       FROM order_t) AS FEEDBACK
       GROUP BY quarter_number ;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,
   ROUND(AVG(CUST_REVIEWS), 2) AS AVG_FEEDBACK
FROM (
   SELECT customer_feedback, quarter_number,
      CASE WHEN customer_feedback = 'Very Bad' THEN 1
      WHEN customer_feedback = 'Bad' THEN 2
      WHEN customer_feedback = 'Okay' THEN 3
      WHEN customer_feedback = 'Good' THEN 4
      WHEN customer_feedback = 'Very Good' THEN 5
   END AS CUST_REVIEWS
   FROM order_t) AS FEEDBACK
   GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	AVG_FEEDBACK
1	3.55
2	3.35
3	2.96
4	2.4

Observations and Insights:

- The average feedback overall is 3.14 which lands in between the ‘Okay’ and ‘Good’ category.
- The average feedback ratings are decreasing over time.
- The highest average feedback was in Q1 with 3.55 (‘Okay’-’Good’) and lowest was in Q4 with 2.4 (‘Bad’-’Okay’).

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT quarter_number,
      ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1
      ELSE 0 END)*100.0/COUNT(*), 2) AS "VERYBAD_PCT",
      ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1
      ELSE 0 END)*100.0/COUNT(*), 2) AS "BAD_PCT",
      ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1
      ELSE 0 END)*100.0/COUNT(*), 2) AS "OKAY_PCT",
      ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1
      ELSE 0 END)*100.0/COUNT(*), 2) AS "GOOD_PCT",
  ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1
    ELSE 0 END)*100.0/COUNT(*), 2) AS "VERYGOOD_PCT"
   FROM order_t
   GROUP BY quarter_number
   ORDER BY quarter_number ASC ;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,
    ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1
    END)*100.0/COUNT(*), 2) AS "VERYBAD_PCT",
    ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1
    END)*100.0/COUNT(*), 2) AS "BAD_PCT",
    ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1
    END)*100.0/COUNT(*), 2) AS "OKAY_PCT",
    ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1
    END)*100.0/COUNT(*), 2) AS "GOOD_PCT",
ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1
  END)*100.0/COUNT(*), 2) AS "VERYGOOD_PCT"
 FROM order_t
 GROUP BY quarter_number
 ORDER BY quarter_number ASC
```

Output:

Showing 4 rows

quarter_number	"VERYBAD_PCT"	"BAD_PCT"	"OKAY_PCT"	"GOOD_PCT"	"VERYGOOD_PCT"
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- The percentage of 'Very Bad' reviews is highest in Q4.
- The percentage of 'Very Good' reviews is highest in Q1.
- The average rating is decreasing over the quarters.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT quarter_number,
     COUNT(order_id)
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number ASC ;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,
    COUNT(order_id) AS NUM_OF_ORDERS
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number ASC
```

Output:

Showing 4 rows

quarter_number	NUM_OF_ORDERS
1	310
2	262
3	229
4	199

Observations and Insights:

- The highest number of orders was in Q1.
- The lowest number of orders was in Q4.
- The number of orders is decreasing overtime, and this could be related to the decreasing average feedback ratings.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query: **A**

```
NET REVENUE:
   1. SELECT
   2. ROUND(SUM(quantity*(vehicle_price-(vehicle_price*(discount/100)
   3. ))), 2)
   4. AS NET_REVENUE
   5. FROM order_t ;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
ROUND(SUM(quantity*(vehicle_price-(vehicle_price*(discount/100)
))), 2)
AS NET_REVENUE
FROM order_t
```

Output:

Showing 1 rows

NET_REVENUE
124714086.32

Solution Query: **B**

```
SELECT quarter_number, NET_REVENUE, PREV_Q_REVENUE,
     ROUND(((NET_REVENUE-PREV_Q_REVENUE)/PREV_Q_REVENUE)
     *100.0, 2) AS PCT_CHANGE
FROM (
SELECT NET_REVENUE, quarter_number,
     LAG(NET_REVENUE) OVER(ORDER BY quarter_number ASC)
     AS PREV_Q_REVENUE
FROM (
SELECT quarter_number,
  ROUND(SUM(quantity*(vehicle_price-(vehicle_price*(discount/100
  )))), 2)
  AS NET_REVENUE
  FROM order_t
  GROUP BY quarter_number)AS LAG_TABLE) AS Q_OVER_Q ;
```

Output:

Output:

Showing 4 rows

quarter_number	NET_REVENUE	PREV_Q_REVENUE	PCT_CHANGE
1	39421580.16		
2	32715830.34	39421580.16	-17.01
3	29229896.19	32715830.34	-10.66
4	23346779.63	29229896.19	-20.13

Observations and Insights:

- The Net revenue of the company was $124,714,086.32
- The percent revenue decreases quarter over quarter.
- The percent change is largest between Q3 and Q4.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT quarter_number,
ROUND(SUM(quantity*(vehicle_price-(vehicle_price*(discount/100)
     ))) ,2) AS NET_REV_PER_Q,
     COUNT(order_id) AS TOTAL_ORDERS_PER_Q
FROM order_t
GROUP BY quarter_number ;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,
ROUND(SUM(quantity*(vehicle_price-(vehicle_price*(discount/100)
    ))) ,2) AS NET_REV_PER_Q,
    COUNT(order_id) AS TOTAL_ORDERS_PER_Q
FROM order_t
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	NET_REV_PER_Q	TOTAL_ORDERS_PER_Q
1	39421580.16	310
2	32715830.34	262
3	29229896.19	229
4	23346779.63	199

Observations and Insights:

- The net revenue decreased from Q1 - Q4.
- The total amount of orders decreased from Q1-Q4.
- The decrease in total orders can explain the lack of net revenue by Q4.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT C.credit_card_type,
      ROUND(AVG(O.discount)*100,0) AS AVG_DISCOUNT
FROM customer_t C JOIN order_t O USING(customer_id)
GROUP BY C.credit_card_type ;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT C.credit_card_type,
    ROUND(AVG(O.discount)*100,0) AS AVG_DISCOUNT
FROM customer_t C JOIN order_t O USING(customer_id)
GROUP BY C.credit_card_type
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	AVG_DISCOUNT
americanexpress	62
bankcard	61
china-unionpay	62
diners-club-carte-blan...	61
diners-club-enroute	60

Observations and Insights:

- The average discount per credit card is similar between them all.
- Laser has the highest average discount of 64%.
- Diners-club-international has the lowest average discount at 58%.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT order_id, order_date, ship_date,
      JULIANDAY(ship_date)-JULIANDAY(order_date) AS DATE_DIFF
FROM order_t
GROUP BY order_id ;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT order_id, order_date, ship_date,
    JULIANDAY(ship_date)-JULIANDAY(order_date) AS DATE_DIFF
FROM order_t
GROUP BY order_id
```

Output:

Showing first 10 rows out of 1000 rows

order_id	order_date	ship_date	DATE_DIFF
0002-7502	2018-11-27	2019-01-01	35
0004-0259	2018-10-28	2019-02-06	101
0006-0078	2018-03-26	2018-05-30	65
0013-2651	2018-10-31	2019-05-06	187
0019-0862	2018-11-10	2019-05-05	176
0049-0032	2018-11-10	2019-05-13	184

Observations and Insights:

- The longest date difference is 525 days for order ID's 33261-460 and 45963-346.
- The shortest date difference is 0 (delivered same day) for the order ID's 51367-016 and 54575-196.
- The long order dates have most likely received bad reviews

Business Metrics Overview

Net Revenue	Total Orders	Total Customers	Average Rating
$124,714,086.32	1000	994	3.14
Last Quarter Net Revenue	Last Quarter Orders	Average Days to Ship	% Good Feedback
$23,346,779.63	199	97.96	39.60%

Business Recommendations: The company's quarter over quarter revenue is rapidly decreasing. To stabilize the revenue, the company could:

- Focus on shortening the average days to ship their product. People are impatient, and if they can get their product to their customers faster, they would generate higher average feedback scores.
- With a higher average feedback score, the company would draw more customers as they would seem more reliable.
- With higher scores, they could market these them to draw in more customers, which will increase their overall net revenue and stabilize it quarter over quarter.,